

SatsPath Demo Flow

This document shows a complete example session using the SatsPath CLI.

Setup

```
cd ~/Desktop/satspath  
cargo build --release
```

Step-by-step

1. Initialize

```
satspath init
```

Output:

```
Created .satspath/registry.json  
Created .satspath/keys.json
```

SatsPath initialized at .satspath/

NOTE: .satspath/ is git-ignored and stays local to this machine.

2. Register

```
satspath register rodrigo@satspath.dev
```

Output:

```
Registered: rodrigo@satspath.dev  
Identity pubkey: 02a1b2c3...  
Fingerprint:    a1b2c3d4
```

Payment methods registered:

- Lightning Address
- Bitcoin on-chain (primary)
- Bitcoin on-chain (secondary, privacy address)
- Ark

Profile signed and stored in .satspath/registry.json

DEMO secret key stored in .satspath/keys.json (git-ignored)

3. Show

```
satspath show rodrigo@satspath.dev
```

Output:

```
Alias:          rodrigo@satspath.dev  
Identity pubkey: 02a1b2c3...  
Fingerprint:    a1b2c3d4  
Signature valid: yes  
Updated at:      1735000000
```

Methods:

- Lightning Address [Lightning]
 Lightning Address: rodrigo@satspath.dev
- Bitcoin (primary) [On-chain]

```
    Address: bc1qa1b2...
- Bitcoin (secondary) [On-chain]
    Address: bc1qb3c4...
- Ark [Ark]
    Server: ark.satspath.dev
```

4. Encode a payment request

```
satspath encode rodrigo@satspath.dev 21000 --memo "coffee"
```

Output:

Encoded SatsPath URI:

```
satspath:v1:eyJ2ZXJzaW9uIjoxLCJhbGhlcyI6InJvZHJpZ29Ac2F0c3BhdGguZGV2IiwiaW1vdW50X3NhdHMiOjIxMDAwLCJtZW1
```

5. Decode

```
satspath decode "satspath:v1:eyJ2ZXJzaW9uIjoxLCJhbGhlcyI6InJvZHJpZ29Ac2F0c3BhdGguZGV2IiwiaW1vdW50X3NhdHMiOjIxMDAwLCJtZW1"
```

Output:

Decoded payment request:

```
Version:      1
Alias:        rodrigo@satspath.dev
Amount:       21000 sats
Memo:         coffee
Profile hint: (none)
```

6. Get a quote

```
satspath quote rodrigo@satspath.dev 21000
```

Output:

Resolving alias 'rodrigo@satspath.dev'...

Verifying signature...

Signature valid.

Checking payment rails for 21000 sats...

Route Quote:

```
Selected rail:  Lightning
Label:         Lightning Address
Reason:        Amount (21000 sats) is below 100000 sats threshold and Lightning is available.
Estimated fee:  2 sats
Confirmation:   instant
```

7. Simulate payment

```
satspath pay rodrigo@satspath.dev 21000
```

Output:

SatsPath Payment Simulation

Resolving alias 'rodrigo@satspath.dev'...

Found profile.

Verifying signed profile...

Signature valid.

Checking available payment rails...

Route selected: Lightning

Selected route: Lightning

Reason: Amount (21000 sats) is below 100000 sats threshold and Lightning is available.

Estimated fee: 2 sats

Executing simulated payment of 21000 sats to rodrigo@satspath.dev...

[Lightning] Generating invoice from rodrigo@satspath.dev...

[Lightning] Invoice received.

[Lightning] Sending payment...

Payment status: simulated_success

DISCLAIMER: This is a simulation. No real Bitcoin was moved.

8. Invite flow for unknown user

```
satspath invite julian@example.com 21000
```

Output:

'julian@example.com' is not registered on SatsPath.

Invite link:

https://satspath.local/claim?alias_hash=a1b2c3d4e5f6g7h8&amount=21000

Alias hash: sha256(julian@example.com)

Amount: 21000 sats

Created at: 1735000000

WARNING: The receiver must claim this payment by generating their own keys locally.

SatsPath never holds or generates keys on behalf of users.

9. Run full demo automatically

```
satspath demo
```

Runs all of the above steps in sequence.